

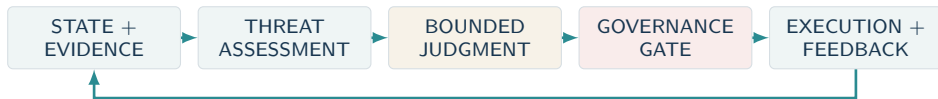
# Autonomous Cybersecurity Operations Center

Optimization with Judgment

A dynamic SOC simulator and its transfer to banking regulation, systemic risk and financial stability

Optimization ranks feasible actions. Judgment determines when changing conditions require priorities, sequences and authority to be reconsidered.

# The central proposition: optimization is not enough



## Optimization asks

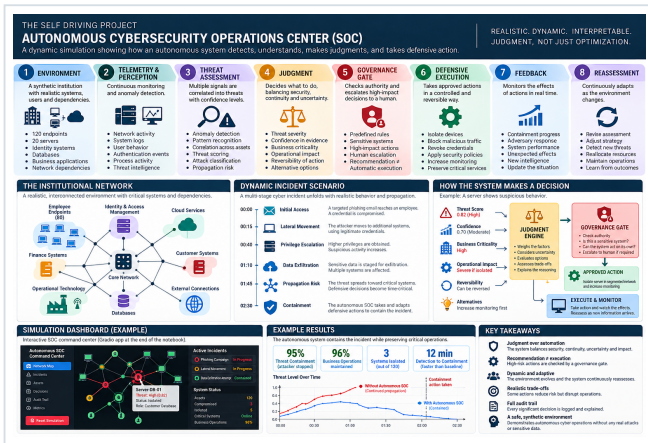
Which feasible defensive action best balances modeled security benefit and operational cost under the current state?

## Judgment asks

Has the environment changed enough that criticality, uncertainty, reversibility or systemic consequences should alter the priority ordering?

**A high threat score is evidence entering a decision. It is not an instruction.**

# The complete autonomous SOC architecture



## Eight layers, one closed loop

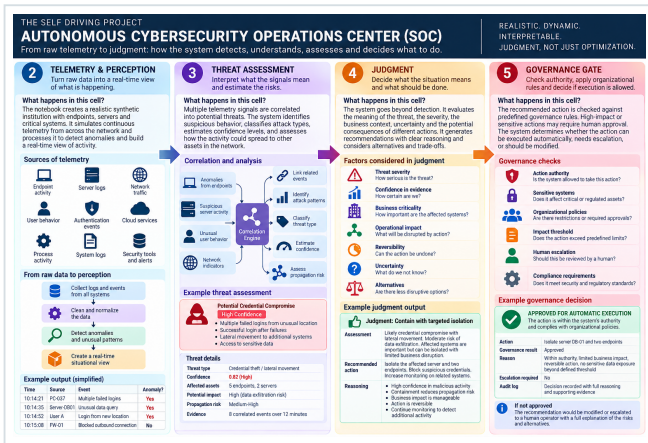
The notebook is deliberately organized as an institutional control system rather than a classifier.

- Environment and telemetry establish the state.
- Assessment converts signals into risk evidence.
- Judgment weighs competing consequences.
- Governance separates recommendation from permission.
- Execution changes the environment.
- Feedback forces reassessment.

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The intellectual center is the transition

# Cells 2-5: build the institution before asking it to decide



## State, network and evidence

The model begins with a synthetic institution: endpoints, servers, critical services and explicit dependencies.

- Telemetry is generated continuously.
- Multiple signals are correlated into evidence.
- Criticality and propagation enter the state representation.
- Governance is already visible before execution.

The system cannot exercise judgment well

if the institutional state is poorly

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# Cells 6–10: a dynamic environment creates the decision problem

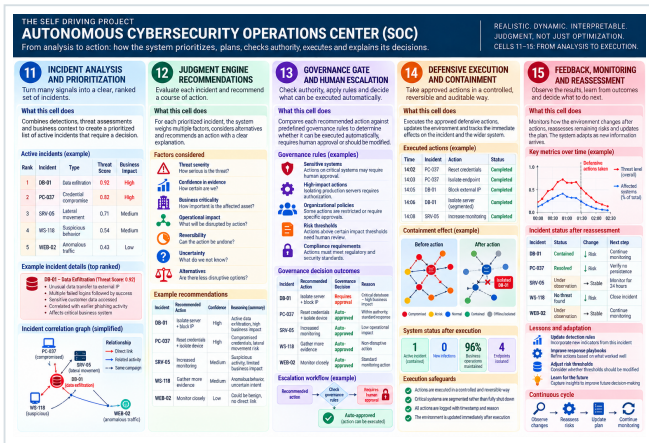


## From environment to response

The simulation evolves through discrete time. Normal activity, ambiguous anomalies and synthetic incidents alter the state.

- Scenarios create changing evidence and propagation pressure.
- The SOC repeatedly detects, assesses and judges.
- Actions are not chosen once; they are reconsidered as the world changes.
- Business continuity remains an explicit competing objective.

# Cells 11–15: judgment and governance become explicit



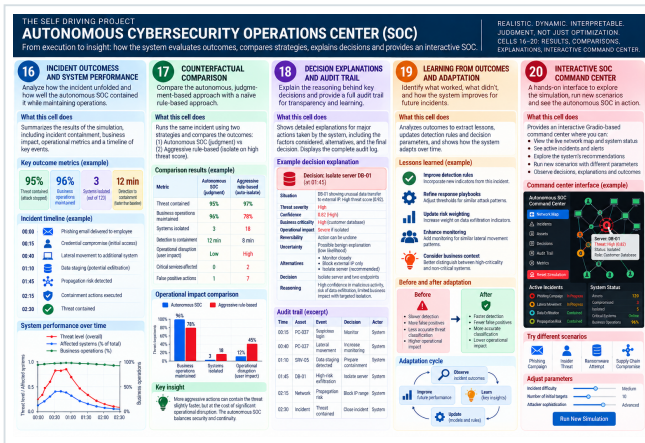
## Recommendation $\neq$ authority

This is the core of the architecture. Candidate actions are compared on more than threat reduction.

- Severity and confidence matter.
- Criticality and operational disruption matter.
- Reversibility and uncertainty matter.
- High-impact actions cross an explicit governance gate.

The machine may recommend an action that it is not authorized to execute.

# Cells 16–20: execution closes the loop

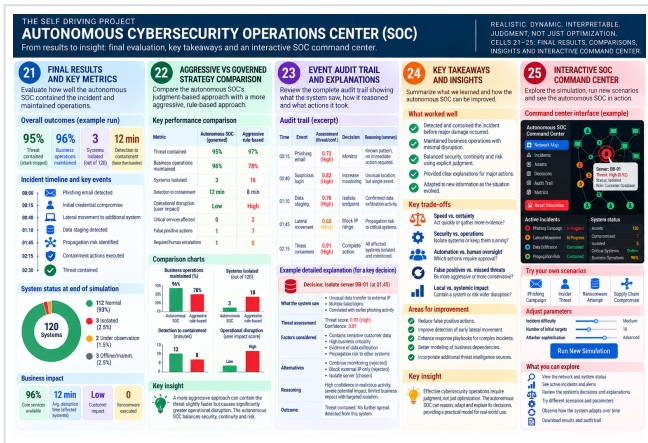


## Act, observe, learn

Approved defensive actions alter the simulated institution, and their consequences become new evidence.

- Execution is bounded and auditable.
- Institutional health is measured after intervention.
- A clean reset prevents stale state from contaminating experiments.
- The next decision uses the new environment, not the old plan.

# Cells 21–25: evaluate whether judgment added value



## Counterfactual, audit and command center

The final cells move from demonstration to evaluation.

► A critical-server dilemma exposes the cost of aggressive automation.

► The integrated experiment is rerunnable from a clean state.

► The audit trail makes consequential choices inspectable.

► A counterfactual compares governed judgment with a more aggressive rule.

► The command center lets the

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user vary evidence and inspect the result

# The general model: optimization with judgment

$$a_t^* = \arg \max_{a \in \mathcal{A}_t} [B(a \mid y_t, p_t) - C(a \mid y_t, p_t)]$$

## 1. Feasible set

Code represents the state, constraints, candidate actions and modeled consequences.

## 2. Priority update

Judgment can alter the relative importance of security, continuity, uncertainty and reversibility.

## 3. Governed action

Delegated authority determines whether the preferred action may execute automatically.

$$p_{t+1} = J(y_t, p_t, z_t) \quad a_t = g(\tilde{a}_t, \text{rules, limits, authority})$$

**The conceptual shift:** the institution does not merely recompute an optimum. It can reconsider which objective deserves precedence as conditions change.

# Why changing priorities change the sequence

**Initial state** several anomalies compete for limited monitoring and intervention capacity

**Initial ordering** investigate the highest expected-risk cases first

**Environmental change** a payment or identity node becomes implicated; propagation pressure rises

**Priority change** systemic importance and irreversibility now outweigh local threat ranking

**Re-sequenced action** preserve critical operations, intensify evidence gathering, escalate high-impact intervention

## The key idea

A sequence is optimal only relative to a state and a priority structure. When either changes, continuing the old sequence can become irrational.

## Judgment enters here

Judgment is the controlled mechanism that recognizes when the ordering of legitimate objectives should change.

# Transfer learning: what transfers from cyber to finance?

## **Asset criticality**

systemically important bank, CCP, payment system

## **Telemetry**

deposits, spreads, haircuts, flows and payments

## **Defensive action**

facilities, restrictions, resolution, intervention

## **Dependency graph**

interbank claims, collateral and funding links

## **Propagation**

runs, fire sales, margin and liquidity spirals

## **Governance**

supervisor, central bank, resolution authority

## **Architectural transfer**

We do **not** transfer a cyber model into banking. We transfer the decomposition of the decision problem: observation, inference, consequence estimation, alternatives, authority, execution and feedback.

**Transfer the architecture of intelligence and control — not the surface variables.**

# Banking regulation as a self-driving supervisory system

**Perception** deposits, wholesale funding, market spreads, collateral, payments, common exposures



**Assessment** idiosyncratic liquidity event or emerging system-wide stress?



**Judgment** when does preserving market discipline become less important than preventing contagion?



**Governance** which authority can provide liquidity, impose restrictions, coordinate or resolve?



**Execution + feedback** intervene, observe flows and prices, then reassess the system

## The supervisory dilemma

A bank can be individually weak without being systemically dangerous. Conversely, a modest local event can become systemic through confidence, funding and network effects.

## Therefore

The relevant objective function can change as the state evolves.

# Replace cyber attacks with speculative attacks

**Signal** withdrawals, short positioning, widening spreads, reserve loss, collateral pressure



**Interpretation** information, noise, coordination, self-fulfilling run or fundamental deterioration?



**Judgment** defend, communicate, provide liquidity, tighten conditions, tolerate adjustment or escalate?



**Strategic feedback** market participants anticipate policy and modify their behavior



**New state** intervention itself changes prices, beliefs, liquidity and the next feasible action set

## The finance extension

Unlike the notebook's synthetic incidents, financial actors may respond strategically to the policy rule itself. Expectations therefore become part of the environment.

This is where bank-run models, global games and market-liquidity feedback become natural extensions of the SOC architecture.

# Other financial applications of optimization with judgment

## **Liquidity crisis management**

sequence facilities, collateral and payment priorities

## **Sovereign / market stress**

intervention versus price discovery and reserve use

## **Recovery and resolution**

preserve critical functions while reallocating losses

## **Central counterparties**

margin, default resources and continuity under stress

## **Credit committees**

restructure, extend, demand collateral or exit

## **Collateral / margin**

re-sequence scarce liquidity as calls and haircuts change

## **Common structure**

Each problem contains scarce resources, changing information, interconnected consequences and multiple legitimate objectives.

## **Common question**

When should a changed environment cause the institution to change not merely the solution, but the priority ordering behind the solution?

# What transfers — and what does not

## Transfers well

- ▶ Separation of state, evidence, feasible actions and priorities.
- ▶ Network criticality and propagation as decision inputs.
- ▶ Explicit comparison of intervention benefit and collateral cost.
- ▶ Governance between recommendation and execution.
- ▶ Closed-loop feedback, provenance and counterfactual evaluation.

## Must be redesigned for finance

- ▶ The variables, causal mechanisms and calibration.
- ▶ Strategic behavior and expectation formation.
- ▶ Data latency, missingness and supervisory reporting structures.
- ▶ Legal mandates and allocation of authority across agencies.
- ▶ The meaning of intervention cost, reversibility and systemic importance.

**The analogy is useful only if we state both the structural continuity and the domain-specific non-equivalence.**

# Governance: judgment must remain bounded discretion



## No invisible authority

A model's confidence does not create a supervisory or central-bank mandate.

## Hard vs. soft constraints

Legal boundaries are not merely another term to trade against a higher score.

## Preserve accountability

Escalation is a designed feature of autonomy, not evidence that autonomy failed.

# The next experiment: a 25-bank systemic-risk simulator

**Environment** 25 banks linked by deposits, interbank claims, securities, collateral and payments

**Shock** withdrawals, asset-price decline, margin calls or speculative pressure

**Perception** liquidity, solvency, spreads, flows and network propagation

**Judgment** distinguish idiosyncratic weakness from systemic danger; reprioritize as contagion evolves

**Governance** central bank, supervisor and resolution authority retain distinct mandates

**Evaluation** compare rule-based intervention with governed judgment under identical shocks

## Research objective

Test whether explicit judgment improves systemic outcomes without turning model discretion into unbounded policy authority.

## Counterfactual

Same shock. Same information.  
Different decision architecture.  
Measure contagion, liquidity, intervention cost and critical-function continuity.

# Five conclusions for institutional design

1. **Optimization and judgment are different forms of intelligence.** Optimization ranks within a specified problem; judgment can reconsider the priority structure.
2. **Changing conditions can change the correct sequence.** The plan is conditional on the state in which it was created.
3. **Recommendation and authority must remain separate.** High-consequence actions require explicit institutional permission.
4. **The architecture transfers naturally to systemic risk.** Financial networks also combine critical nodes, propagation, uncertainty and costly interventions.
5. **Autonomy is a closed-loop institutional capability.** Perceive, assess, judge, govern, execute, observe and revise.

# Optimization with Judgment

The autonomous institution does not merely react faster. It knows when a changed environment requires priorities, sequences and authority to be reconsidered.

Perception → assessment → judgment → governance → execution → feedback